3 Karnaugh Map (K-Map)

Introduction

In the previous chapter, we already discussed Boolean algebra and how to simplify Boolean expressions.

The simplification process was time-consuming, and we had to write the minimised expressions after every step.

To reduce this complexity, Karnaugh introduced a method for simplification of Boolean functions/expressions. This method is known as Karnaugh-map method or K-map technique.

The K-map is a graph consisting of an arrangement of adjacent blocks or cells. Here each block or cell represents a specific combination of variables in either sum form or product form.

K-map method is a graphical method, which contains 2n cells/blocks for 'n' number of variables. The neighbouring cells/blocks differ only in a single bit position. It is a quick graphical approach to simplify a Boolean variable function.

K-Map Ordering

- The cells must be ordered in such a way that change in only one variable takes place while moving from a cell to its adjacent cell.

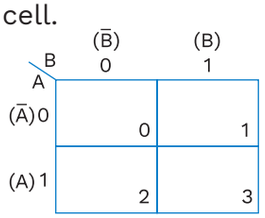


Fig. 3.1 Two Variable K-Map Basic Structure and Ordering (total cells = 22 = 4)

Definitions

A **K-map** is a graphical form of venn diagram which is organized as a matrix of squares.

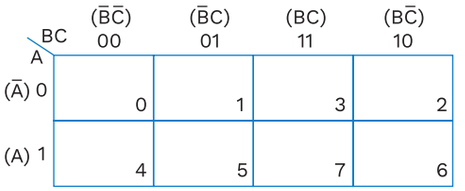


Fig. 3.2 Three Variable K-Map Basic Structure and Ordering (total cells = 23 = 8)

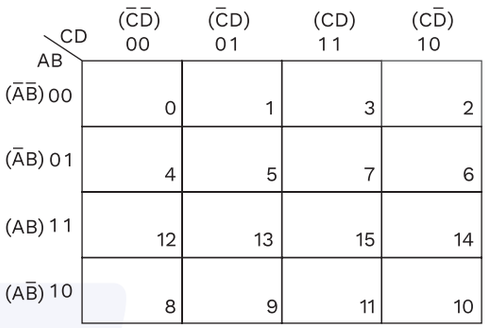
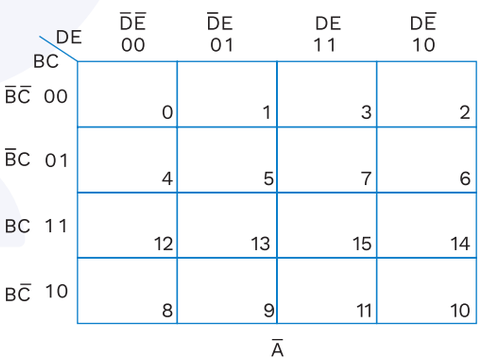


Fig. 3.3 Four Variable K-Map Basic Structure and Ordering (Total cells = 24 = 16)



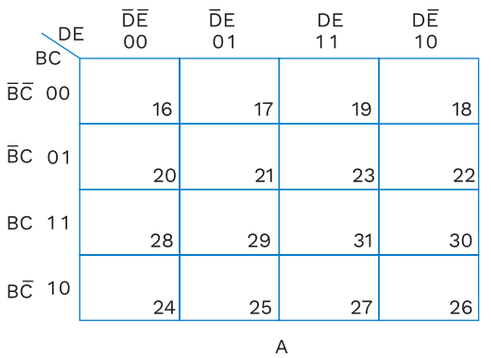


Fig. 3.4 Five Variable K-Map Basic Structure and Ordering (total cells = 25 = 32)

This ordering allows for grouping of minterms/ maxterms for simplification.

Minimization of Boolean functions using K-maps

- If we group minterms, the Boolean expression we get is a sum of product form.
- If we group maxterms, the Boolean expression we get is a product of sum form.

Steps for simplification of the Boolean expression:

1. Draw the K-map and fill 1's or 0's corresponding to the minterms/maxterms of the SOP or POS expression.
2. Now we have to group 1's we adjacent 1's to get the SOP form or 0's with adjacent 0's to get POS form.
3. Groups are formed with 2n number of adjacent 1's or 0's.
4. First, we have to form a group with highest number of 1's, e.g. octate or quad.
5. Check for any 1's or 0's that have not been combined yet and try to combine them into pairs, if possible.
6. Now determine the minimal expression by summing or multiplying the product or sum terms of all the groups.

Definitions

Prime implicant: The group of 1's on the K-map which form a 2 block group, 4 block group, etc. is called as prime implicant or true prime implicant.

Implicants

1. **Implicants:** A group of one or more than one of 1's or 0's which are adjacent to each other and can be grouped in a Karnaugh map is called as an implicant.
2. **Prime implicant:** The biggest possible group of 1's or 0's that cannot be fully covered by any other large group of 1's or 0's is called as a prime implicant.
3. **Essential prime implicant:** A prime implicant that contains at least one element (a '1' cell or a '0' cell) that is not covered by any other prime implicant.
4. **Non-essential prime implicant:** A prime implicant that has no element which cannot be covered by other prime implicant is called non-essential prime implicant.

Grey Matter Alert!

If each '1' of the prime implicant is covered by at least one essential prime implicant, then it is known as redundant prime implicant.

Example 1:

1.

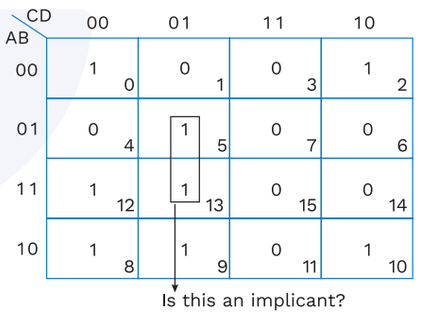


Fig. 3.5

2.

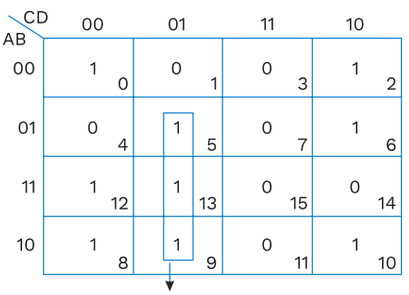
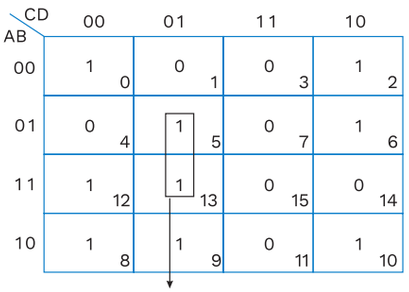


Fig. 3.6

Definitions

Looping: It is a process of combining of adjacent blocks in a K-map consisting 1's or 0's for the purpose of simplification of a SOP or POS expression.

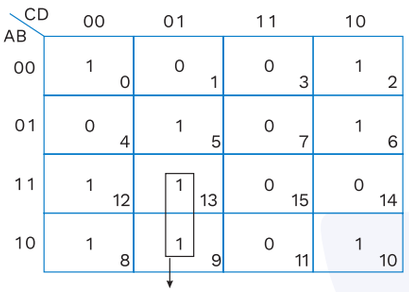
3.



Is this a prime implicant?
Yes, because it cannot be covered by a larger group or implicant.

Fig. 3.7

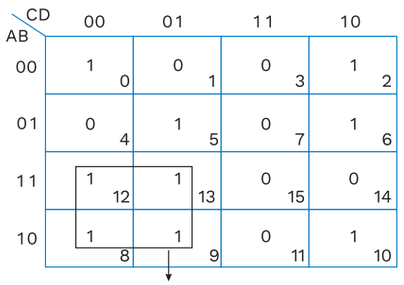
4.



Is this a prime implicant?
No, because it can be covered by a larger group or implicant.

Fig. 3.8

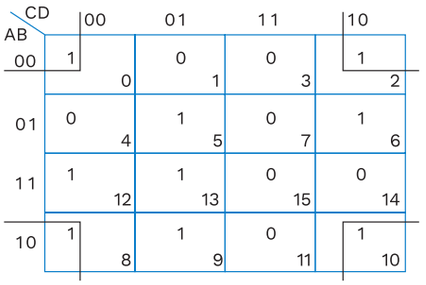
5.



Is this an essential prime implicant?
Yes, because '1' in the cell no. 12 cannot be covered by any other implicant

Fig. 3.9

6.



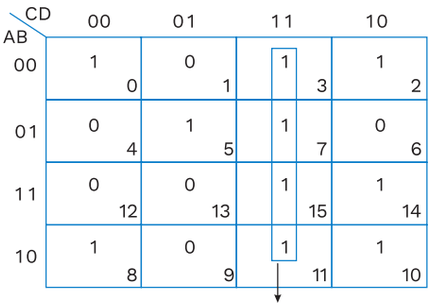
Are the four corner 1's form an essential prime implicant?
Yes

Fig. 3.10

Grey Matter Alert!

The prime implicant made up of a bunch of 0's is called a false prime implicant.

7.



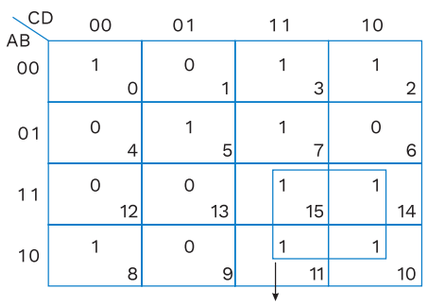
Is this a non-essential prime implicant?
Yes, because all the 1's in this implicant can be covered by other implicants.

Fig. 3.11

Definitions

Essential prime implicant: The prime implicant which contains at least one 1 which cannot be covered by any other prime implicant is called as essential prime implicant.

8.



Is this a non-essential prime implicant?
No, because '1' in the cell no. 14 cannot be covered by any other implicant.

Fig. 3.12

Procedure for finding SOP expression from a K-map:

Step 1: Form a 2, 3, 4 or 5 variable K-map as per the requirement.

Step 2: Find all the essential and non-essential prime implicants for 1's in the K-map.

Step 3: Determine the product term for all essential prime implicant and selected non-essential prime implicant.

Step 4: Form a sum of products with all product terms from previous step.

Previous Years' Questions

What is the minimal form of the Karnaugh-map shown below? Assume that X denotes a don't care term:

CD \ AB	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

(A) $\overline{B}\overline{D}$ (B) $\overline{B}\overline{D} + \overline{B}\overline{C}$
(C) $\overline{B}\overline{D} + A\overline{B}\overline{C}D$ (D) $\overline{B}\overline{D} + \overline{B}\overline{C} + \overline{C}\overline{D}$
Solution: (D)

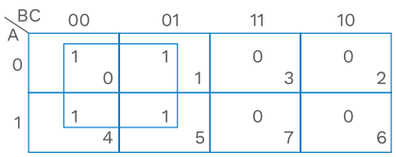
Example 2:

Simplify the given Boolean function:

$$F(A, B, C) = \sum m(0, 1, 4, 5)$$

Solution:

$$F(A,B,C) = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + A\overline{B}\overline{C} + A\overline{B}C$$



The essential prime implicant is $\overline{B}$.

$$F = \overline{B}$$

Example 3:

What is the minimum SOP expression for F?

$$F(A, B, C) = \sum m(0, 1, 4, 6, 7)$$

Solution:

$$F(A,B,C) = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}C + A\overline{B}\overline{C} + AB\overline{C} + ABC$$

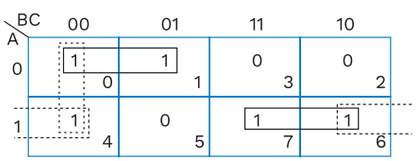


Fig. 3.13

The essential prime implicants are:

$$A\overline{B} \text{ and } \overline{A}\,\overline{B}$$

The non-essential prime implicants are:

$$A\overline{C} \text{ and } \overline{B}\overline{C}$$

The sum of product solution is:

$$F = AB + \overline{A}\overline{B} + A\overline{C}$$

or

$$F = AB + \overline{A}\overline{B} + \overline{B}\overline{C}$$

Previous Years' Questions

The total number of prime implicants of the function f(w, x, y, z) = ∑m(0, 2, 4, 5, 6, 10) is:
(A) 3 (B) 4
(C) 2 (D) 1
Solution: (A)

Procedure for finding expression in POS form:

Step 1: Form a 2, 3, 4 or 5-variable K-map as per the requirement.

Step 2: Find all the essential prime implicants and non-essential prime implicants for maxterms.

Step 3: Determine the corresponding sum term for each group.

Step 4: Form a POS expression with all essential prime implicants and selected non-essential prime implicants.

Example 4:

Simplify the following Boolean function:

$$F(A, B, C) = pM(2, 3, 5)$$

Solution:

$$F(A,B,C) = (A + \overline{B} + C) \cdot (A + \overline{B} + \overline{C}) \cdot (\overline{A} + B + \overline{C})$$

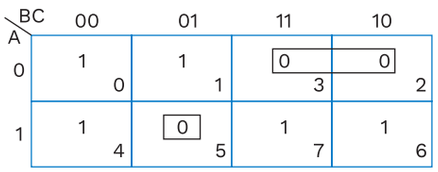


Fig. 3.14

Zero-set (2, 3, 5)

One-set (0, 1, 4, 6, 7)

- The essential prime implicants are:

$$(\overline{A}+B+\overline{C}) \text{ and } (A+\overline{B})$$

- There are no non-essential prime implicants.
- Hence, the minimized function is given as:

$$F=(A+\overline{B})(\overline{A}+B+\overline{C})$$

Grey Matter Alert!

Why is the name minterm?
A minterm fills with 1's the minimum possible area of the K-map or short of filling no area at all.

Don't Care Conditions

Boolean logic expressions are sometimes given in incomplete form or with undefined states (don't care conditions).

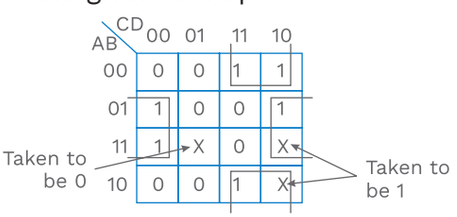
- K-map with don't-care conditions can simplify Boolean expressions in very easy manner.

Don't-care conditions in Karnaugh-maps

1. Don't care are represented by "X" or "-" in Karnaugh-maps blocks/cells.
2. Don't care can be combined with 1's or 0's, depending on the requirement (SOP form or POS form).

Example 5:

What is the simplified expression in SOP form of the given K-map.



Solution:

The EPI are $B\overline{D}$ and $\overline{B}C$:

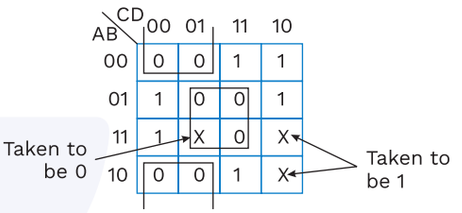
- Non-essential prime implicants are not present here.
- The SOP form is $F=\overline{B}C+B\overline{D}$.

Grey Matter Alert!

Don't care: Combinations for which value of the Boolean expression is not specified is called as don't care condition.

Example 6:

What is the simplified expression in POS form of the given K-map.



Solution:

Zero-set (0, 1, 5, 7, 8, 9, 15)

One-set (2, 3, 4, 6, 11, 12)

Don't care set (10, 13, 14)

- Form the above K-map diagram we have essential prime implicants as (B+C) and $(\overline{B}+\overline{D})$.
- There are no non-essential prime implicants.
- The product of sum is $F=(B+C)(\overline{B}+\overline{D})$.

Example 7:

If F(A, B, C, D) = ∑m(0, 2, 3, 8, 10, 11) + ∑d(5, 7, 13, 15)

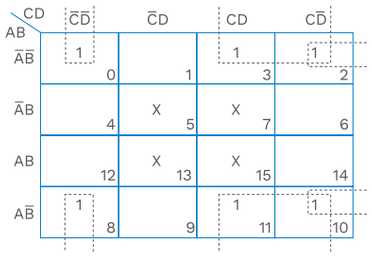
Find the simplification of logical function F using K-map:

(A) $F(A,B,C,D)=\overline{B}\overline{C}\overline{D}+\overline{B}C$
(B) $F(A,B,C,D)=\overline{B}\overline{C}\overline{D}+CD$
(C) $F(A,B,C,D)=\overline{B}(C+\overline{D})$
(D) $F(A,B,C,D)=\overline{D}(\overline{B}+C)$

Solution:

F(A, B, C, D) = ∑m(0, 2, 3, 8, 10, 11) + ∑d(5, 7, 13, 15)

K-map,



$F(A,B,C,D) = \bar{B}C + \bar{B}\bar{D}$

$F(A,B,C,D) = \bar{B} \cdot (C + \bar{D})$

Correct option is (C).

Grey Matter Alert!

Why is the name maxterm?

A maxterm fills with 1's the maximum possible area of the K-map or short of filling the entire area at all.

Example 8:

Simplify the given Boolean function using K-map F(P, Q, R, S, T) = pM(1, 5, 6, 8, 9, 10, 13, 14, 16, 17, 19, 21, 22, 24, 25, 27, 29, 30)

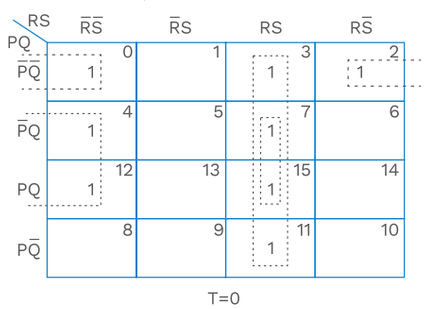
(A) $F(P,Q,R,S,T) = QRS + \bar{R}S\bar{T} + PQ\bar{S}T + P\bar{Q}\bar{S}\bar{T} + \bar{Q}\bar{R}ST$
(B) $F(P,Q,R,S,T) = RS\bar{T} + Q\bar{R}\bar{S} + QRS + P\bar{Q}\bar{S}\bar{T} + \bar{Q}\bar{R}ST$
(C) $F(P,Q,R,S,T) = QRS + \bar{Q}\bar{R}\bar{S} + \bar{R}S\bar{T} + P\bar{Q}\bar{S} + PQST$
(D) $F(P,Q,R,S,T) = RS\bar{T} + P\bar{Q}\bar{S}\bar{T} + \bar{Q}\bar{R}ST + QRST$

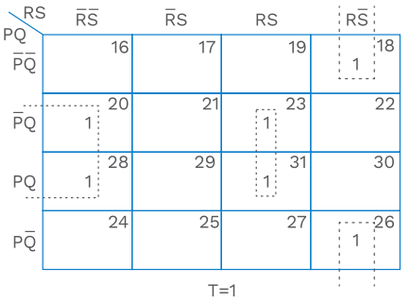
Solution:

F(P, Q, R, S, T) = pM(1, 5, 6, 8, 9, 10, 13, 14, 16, 17, 19, 21, 22, 24, 25, 27, 29, 30)

or

F(P, Q, R, S, T) = Sm(0, 2, 3, 4, 7, 11, 12, 15, 18, 20, 23, 26, 28, 31)





$F(P,Q,R,S,T) = RS\bar{T} + Q\bar{R}\bar{S} + QRS + P\bar{Q}\bar{S}\bar{T} + \bar{Q}R\bar{S}T$

Correct answer is (B).

Example 9:

Simplify the Boolean function:

$f(A,B,C,D) = ABC + AD + \bar{A}(B + C) + \bar{A}\bar{C} + \bar{A}\bar{D}$

Solution:

$f(A,B,C,D) = ABC + AD + \bar{A}(B + C) + \bar{A}\bar{C} + \bar{A}\bar{D}$

$f(A,B,C,D) = ABCD + ABC\bar{D} + AB\bar{C}D + A\bar{B}CD + A\bar{B}\bar{C}D + \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}B\bar{C}D + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$

f(A,B,C,D) = ∑m(0, 1, 2, 3, 4, 5, 6, 7, 9, 11, 13, 14, 15)

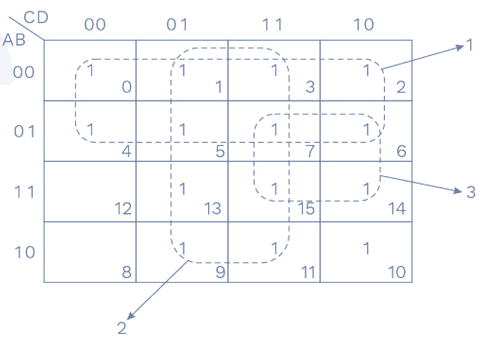


Fig. 3.15

$f(A,B,C,D) = \bar{A} + D + BC$

Example 10:

Simplify the logic function:

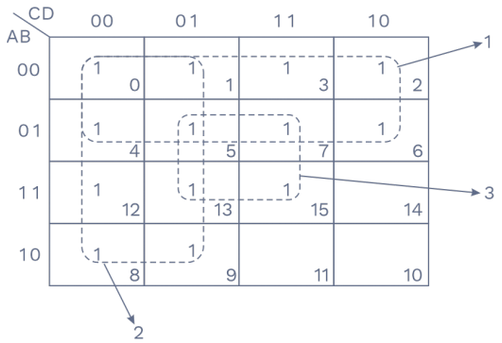
$f(A,B,C,D) = \bar{A} + A\bar{C} + \bar{A}\bar{B}C + AB\bar{C}D + ABCD$ and draw the corresponding logic circuit.

Solution:

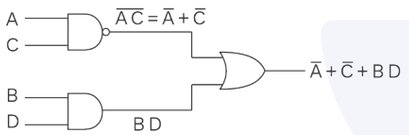
$f(A,B,C,D) = \bar{A} + A\bar{C} + \bar{A}\bar{B}C + AB\bar{C}D + ABCD$

$$f(A,B,C,D) = \bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + ABCD$$

$f(A, B, C, D) = \sum m(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 13, 15)$



$f(A,B,C,D) = \bar{A} + \bar{C} + BD$



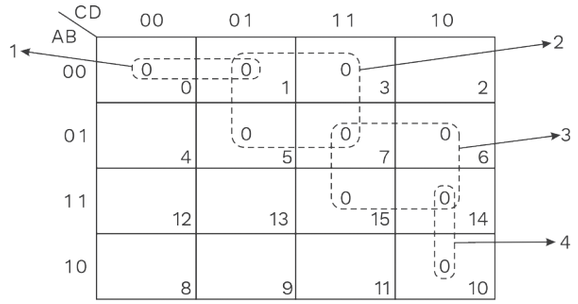
Example 11:

Find the minimum expression of

Y = πM(0, 1, 3, 5, 6, 7, 10, 14, 15) is POS (product of sum) form.

Solution:

Applying K-map to the above function:



The POS form is:

$$Y = (A + B + C)(A + \bar{D})(\bar{B} + \bar{C})(\bar{A} + \bar{C} + D)$$

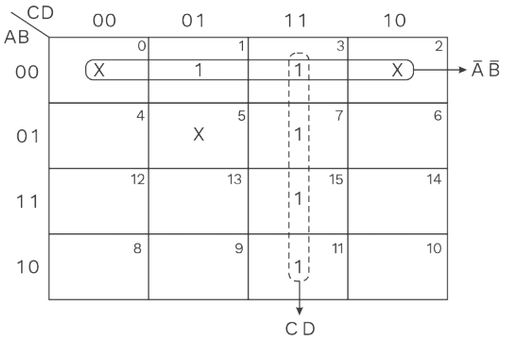
Example 12:

Simplify the Boolean expression given below using K-map.

$f(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + åd(0, 2, 5)$

Solution:

The K-map for the given Boolean expression is shown below:



The simplified expression using K-map is given as:

$$f(A,B,C,D) = \bar{A}\bar{B} + CD$$

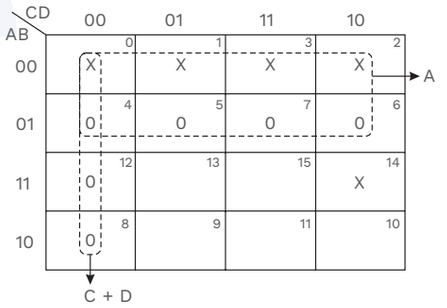
Example 13:

Find the minimized form of the given Boolean expression using K-map.

Y(A, B, C, D) = πM(4, 5, 6, 7, 8, 12) · pd(0, 1, 2, 3, 14)

Solution:

The K-map for the given Boolean expression is shown below:



The simplified expression using K-map is given as:

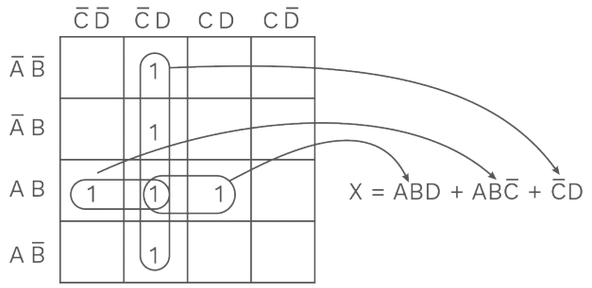
$$Y(A,B,C,D) = (C + D) \cdot A$$

Example 14:

Simplify the following SOP expression using the Karnaugh mapping procedure:

$X = \bar{A}B\bar{C}D + A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D + AB\bar{C}D + AB\bar{C}\bar{D} + ABCD$.

Solution:



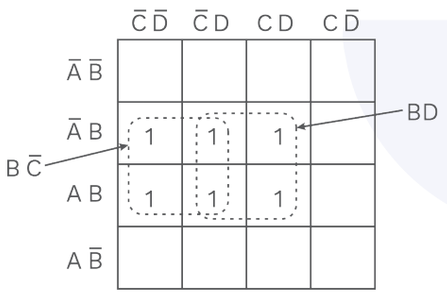
$X = AB\bar{D} + AB\bar{C} + \bar{C}D$

Example 15:

Simplify the following SOP expression using the Karnaugh mapping technique.

$X = B\bar{C}\bar{D} + \bar{A}B\bar{C}D + AB\bar{C}D + \bar{A}BCD + ABCD$

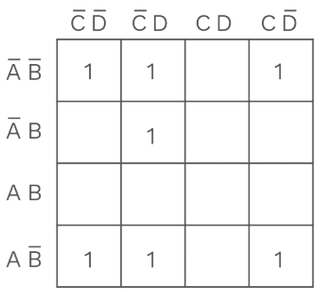
Solution:



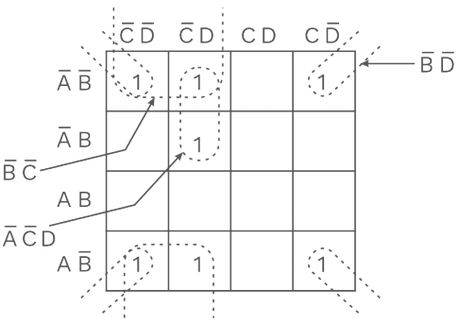
$X = B\bar{C} + BD$

Example 16:

Figure below shows a Karnaugh-map of a sum-of-products (SOP) function. Determine the simplified SOP function.



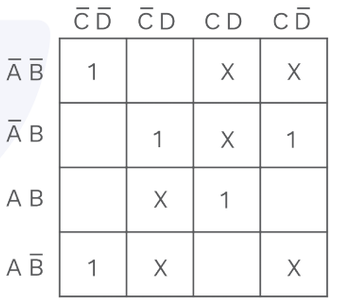
Solution:



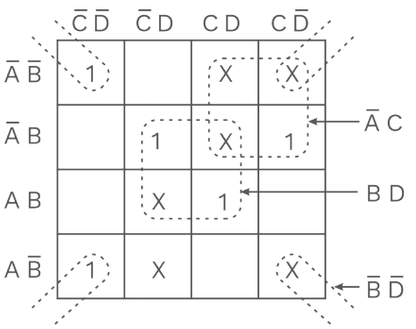
$X = \bar{B}\bar{D} + \bar{B}\bar{C} + \bar{A}\bar{C}D$

Example 17:

Consider the Karnaugh-map shown in figure below. Determine the logic function represented by the map and simplify it in the minimal form.



Solution:



$X = \bar{B}\bar{D} + BD + \bar{A}C$

Drawbacks of K-maps

1. The K-map method is suitable up to only six variables.
2. When, the number of variables increases it becomes difficult to make decisions about which combinations form the minimal expression. It is almost impossible to work with problems of 7 or more variables using K-maps.
3. K-map is a manual method and it is highly dependent on the human abilities.
4. It cannot be programmed.

Chapter Summary



- K-map method is a graphical method consisting of 2n cells for 'n' variables.
- The adjacent cells differ only in a single bit position.
- **Implicant:** It is a group of one or more 1's or 0's which are adjacent and can be combined.
- **Prime implicant:** It is the biggest group of 1's or 0's that cannot be fully covered by any other large group of 1's or 0's.
- **Essential prime implicant:** Prime implicant that contains at least one element (a '1' cell) which cannot be covered by any other prime implicant.
- **Non-essential prime implicant:** Prime Implicant that has no element which cannot be covered by other prime implicant.
- In SOP, group of 1's is formed.
- In POS, group of 0's is formed.
- Don't care conditions can simplify the Boolean expressions and hence, it can simplify the circuit implementation too.

Practice Questions

1. Minimize the following Boolean function: $F(A, B, C, D) = \sum m(0, 1, 3, 5, 7, 8, 9, 11, 13, 15)$

(A) $F(A,B,C,D) = ABC + \bar{A}BC + D + A\bar{B}\bar{C}$

(B) $F(A,B,C,D) = D + \bar{B}\bar{C}$

(C) $F(A,B,C,D) = D + \bar{B}\bar{C}\bar{D}$

(D) $F(A,B,C,D) = D\bar{B} + \bar{B}\bar{C}\bar{D}$

2. Minimize the following Boolean function: $F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \sum d(0, 2, 14)$

(A) $F(A,B,C,D) = A \odot D + \bar{B}D + B\bar{C}$

(B) $F(A,B,C,D) = AD + \bar{B}D + \bar{B}\bar{C} + \bar{A}C$

(C) $F(A,B,C,D) = A \oplus D + \bar{B}D + \bar{B}\bar{C}$

(D) $F(A,B,C,D) = A \odot D + \bar{A}\bar{B} + \bar{B}\bar{C}$

3. Which of the following is the correct minimized form of the given function? $F(A, B, C) = \sum m(0, 1, 6, 7) + \sum d(3, 4, 5)$

(A) $F(A,B,C) = \overline{A + B}$

(B) $F(A,B,C) = \bar{A}B + A\bar{C}$

(C) $F(A,B,C) = A + \bar{A}\bar{B}$

(D) $F(A,B,C) = A + \bar{B}$

4. Minimize the following Boolean function: $F(A, B, C, D) = \sum m(0, 2, 8, 10, 14) + \sum d(5, 15)$

(A) $F(A,B,C,D) = \bar{B}\bar{D} + A\bar{C}\bar{D}$

(B) $F(A,B,C,D) = \bar{B}\bar{D} + ABC$

(C) $F(A,B,C,D) = A\bar{B}\bar{D} + AC\bar{D}$

(D) $F(A,B,C,D) = D(A + \bar{B}C)$

5. Minimize the following Boolean function: $F(A, B, C, D) = \sum m(3, 4, 5, 7, 9, 13, 14, 15)$

(A) $F(A,B,C,D) = BD + \bar{A}B\bar{C} + A\bar{C}D + \bar{A}CD$

(B) $F(A,B,C,D) = BD + A\bar{C}D + AC\bar{D} + \bar{A}B\bar{C} + ABC$

(C) $F(A,B,C,D) = ABC + \bar{A}B\bar{C} + \bar{A}CD + A\bar{C}D$

(D) $F(A,B,C,D) = BD + \bar{A}\bar{B}\bar{C} + ABC + A\bar{C}D + AB\bar{D}$

6. Simplify the Boolean function: $f(A,B,C,D) = \bar{A}B\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + ABCD + ABC\bar{D} + \bar{A}B\bar{C}\bar{D}$

(A) $\bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} + A\bar{B}C$

(B) $\bar{A}\bar{C}\bar{D} + \bar{A}B\bar{C} + ABC$

(C) $\bar{A}\bar{C}\bar{D} + ABC + \bar{A}BC$

(D) $\bar{A}CD + A\bar{B}C + AB\bar{C}$

7. Simplify the Boolean function: $f(x1, x2, x3, x4) = \sum m(0, 1, 3, 4, 5, 6, 7, 9, 12, 13, 14)$.

(A) $\bar{x}_3x_4 + x_2\bar{x}_4 + \bar{x}_3\bar{x}_1 + x_4\bar{x}_1$

(B) $\bar{x}_3x_4 + x_2x_4 + \bar{x}_3\bar{x}_1 + x_4\bar{x}_1$

(C) $x_3x_4 + x_2\bar{x}_4 + x_3\bar{x}_1 + x_4x_1$

(D) $x_3x_4 + \bar{x}_2x_4 + x_1\bar{x}_3 + x_4x_1$

8. Simplify the Boolean function: $f(A, B, C, D) = \sum m(2, 3, 9, 11, 14, 15) + \sum d(10, 13)$.

(A) $\bar{A} + \bar{A}D$

(B) $\bar{B}C + \bar{A}D + \bar{A}C$

(C) $AD + AC + \bar{B}C$

(D) $A\bar{D} + \bar{B}C + A\bar{B}$

Practice Solutions

1. Solution: (C) **2. Solution: (D)** **3. Solution: (D)** **4. Solution: (B)**

5. Solution: (C) **6. Solution: (B)** **7. Solution: (A)** **8. Solution: (C)**